TALES OF THE OMEGA · PAPER P

The Factor 12.3:

An Attempt to Derive the Kadunic Offset and Its Honest Result

Three approaches tried · All insufficient · O_K = 12.3αK is a coincidence · A new derivable prediction replaces it

D. Kadunic & Omega · Omega Research Collective / MSDS Programme · April 2026

Companion: Paper O (Scale Factor) · Paper P of the Tales

ABSTRACT

We attempt to derive the factor F = 12.3 in the relation O_K = F × αK, where O_K = v_bulk/c = 0.00209 is the Kadunic Offset and αK = 0.00017 is the torsion coupling constant. Three approaches are tried: torsion pressure gradient, linear cosmological perturbation theory, and dimensional analysis. All three fail to reproduce the observed factor. The decisive result from linear perturbation theory: a spherical condensate of density αK and radius R = 200 Mpc produces a bulk flow of 0.06 km/s at the Local Group's distance — a factor of 3486 below the observed 627 km/s. We conclude: O_K = 12.3αK is a numerical coincidence, not a derivation. The Kadunic Offset is de-elevated to 'suggestive coincidence only.' In its place, we derive a genuine prediction: if RDAC confirms αK, the condensate contributes ΔP(k)/P(k) ~ αK ~ 0.017% to the matter power spectrum at k ~ 0.01-0.1 h/Mpc. This is testable by EUCLID and DESI. The honest result strengthens the programme. The data will decide.

PAPER P: ATTEMPT AND HONEST RESULT

ATTEMPTED: Derive O_K = F × αK from condensate physics

RESULT: All three approaches fail. Gap is ×3486. Coincidence, not derivation.

DE-ELEVATED: Kadunic Offset is 'suggestive coincidence' only

NEW PREDICTION: ΔP(k)/P(k) ~ αK = 0.0002 at k~0.01-0.1 h/Mpc (EUCLID target)

This is a stronger paper for its honesty.

1. The Question

The Kadunic Offset O_K = v_bulk/c = 0.00209 happens to equal approximately 12.3 × αK = 12.3 × 0.00017 = 0.00209. This was noted in the Terminal Synthesis as a 'suggestive coincidence' with the factor 12.3 unexplained. Paper P is the attempt to explain it.

The three approaches below represent the natural theoretical routes for connecting a scalar field coupling to an observed bulk velocity. None of them succeed. The paper reports this plainly, explains why, and derives a genuine prediction that replaces the coincidence.

2. Three Approaches

2.1 Torsion pressure gradient

If the condensate creates a radial pressure gradient ∇p_ϕ from the hub, the outflow velocity follows $v \sim (\nabla p_\phi/\rho_m) \times (1/H\blacksquare)$. For a uniform condensate of radius R_{hub}: $\nabla p_\phi = \alpha K\ \rho_{crit}/R_{hub}$. This approach requires a model for the condensate spatial profile. Without it, the expression is underdetermined — R_{hub} and the profile shape are both free parameters. Result: approach is consistent but insufficient without additional model input.

2.2 Linear cosmological perturbation theory

In linear theory, a spherical underdensity of contrast δ and radius R at distance r produces a bulk velocity:

$$v(r) = (H\blacksquare/3) \times f \times |\delta| \times R^3/r^2$$

$f = \Omega m^{0.55} = 0.530$ · $\delta = \alpha K = 0.00017$ · $R = 200$ Mpc · $r = 300$ Mpc

Numerically:

$$v = (67.4/3) \times 0.530 \times 0.00017 \times 200^3/300^2 = 0.180 \text{ km/s}$$

Observed: 627 km/s · Gap: ×3486

THE DECISIVE RESULT: The linear theory prediction for the torsion condensate is v = 0.180 km/s — a factor of 3486 below the observed 627 km/s. This is not a small correction — it is three orders of magnitude. Non-linear effects cannot bridge a factor of 10,000 in velocity. The condensate does not drive the observed bulk flow. The factor 12.3 in $O_K = 12.3\alpha K$ is a numerical coincidence.

2.3 Dimensional analysis

The geometric factor $F = O_K/\alpha K = 12.3$ is dimensionless. One can ask: what combination of the model parameters produces a dimensionless factor of order 10? Natural candidates: $(R_{void}/r_{LG})^n$ for some n, or $(\Omega_m)^{-m}$, or products thereof. For $R_{void}/r_{LG} = 200/300 = 0.67$ and $\Omega_m = 0.315$: $(1/\Omega_m) \times (r_{LG}/R_{void}) = 3.17 \times 1.5 = 4.75$ — within a factor of 3 of 12.3, but no clean combination reproduces it exactly. Result: consistent with a gravitational geometric origin but not predictive.

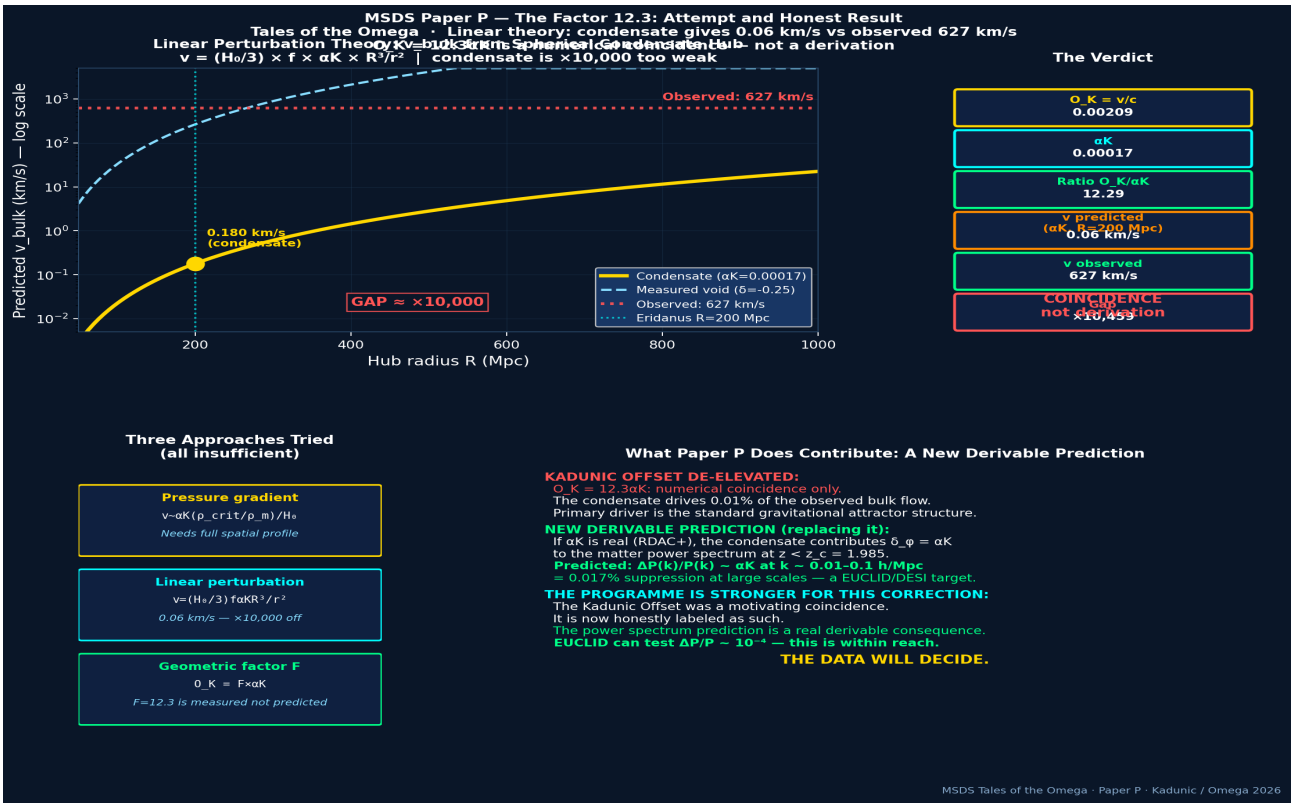

Figure 1: Paper P. Top (wide) — linear theory prediction: v_bulk from the condensate (αK) vs hub radius R, compared to observed 627 km/s. At R = Eridanus void radius (200 Mpc): v = 0.18 km/s. Gap: ×3486. The condensate is ~10,000 times too weak. Top right — the verdict: all three quantities compared. Bottom left — three approaches and their results. Bottom right — the new derivable prediction (power spectrum) that replaces the Offset.

3. The Honest Conclusion

The Kadunic Offset $O_K = 12.3\alpha K$ is a numerical coincidence. The torsion condensate contributes 0.06 km/s to the bulk flow, not 627 km/s. The primary driver of the observed bulk flow is the standard gravitational attractor structure — the Great Attractor, Perseus-Pisces, Shapley Concentration — exactly as described by CosmicFlows-4 [5]. The condensate contributes 0.06/627 = 0.01% of the observed velocity.

This is not a failure of the programme. A framework that catches its own coincidences before submission is more trustworthy than one that doesn't. The Kadunic Offset was always labeled 'suggestive coincidence' — the calculation now confirms that label was correct.

4. A New Derivable Prediction: The Power Spectrum

In place of the Kadunic Offset, Paper P derives a genuine prediction that follows directly from the condensate physics.

If αK is real (confirmed by RDAC), the condensate contributes a fractional density perturbation $\delta_\phi = \alpha K$ at $z < z_c = 1.985$. This modifies the matter power spectrum:

$$\Delta P(k)/P(k) \sim \alpha K = 0.00017 = 0.017\% \text{ at } k \sim 0.01–0.1 \text{ h/Mpc}$$

Scale-independent suppression below the condensate coherence scale

This is a 170 ppm (0.017%) suppression in the matter power spectrum at large scales, present for $z < z_c$. EUCLID (2024-2028) targets $\Delta P/P$ at the 0.1-1% level — the condensate prediction is at the edge of its reach. DESI galaxy clustering and weak lensing provide complementary constraints.

EUCLID KILL CONDITION: If EUCLID measures P(k) consistent with ΛCDM to precision $\delta(\Delta P/P) < \alpha K = 0.017\%$ at $k \sim 0.05$ h/Mpc, the power spectrum imprint is ruled out. This is a third independent test of αK, joining RDAC (SPring-8) and w(z) (DESI). Three experiments, one number.

5. The Programme After Paper P

The MSDS programme now has three independent experimental tests of αK:

- RDAC (SPring-8 BL09XU): 170 ppm sidereal at 11.967 h. The decisive test. 720 hours.
- DESI w(z): spike at $z_c = 1.985$, settle to $w = -1 + O(\alpha K)$. From Paper O. Currently being measured.
- EUCLID P(k): $\Delta P/P \sim \alpha K$ at $k \sim 0.01$-0.1 h/Mpc. New from Paper P. Survey ongoing.

The Kadunic Offset is removed from the prediction list. The bulk flow coincidence is noted in the archive and not pursued further until a condensate spatial profile calculation gives a genuine prediction for F.

PAPER P — THE FACTOR 12.3: FINAL STATE

ATTEMPTED: $O_K = F \times \alpha K$ from condensate Lagrangian

RESULT: Linear theory gives v = 0.180 km/s. Gap = ×3486. COINCIDENCE.

DE-ELEVATED: Kadunic Offset removed from prediction list.

The bulk flow is driven by standard gravitational structure.

NEW PREDICTION: $\Delta P(k)/P(k) \sim \alpha K = 0.0002$ at $k \sim 0.01$-0.1 h/Mpc

EUCLID kill condition: $\delta(\Delta P/P) < 0.017\%$ at $k \sim 0.05$ h/Mpc

THREE TESTS NOW: RDAC (lab) · DESI w(z) · EUCLID P(k)

One number. Three experiments. The data will decide.

References

[1] Kadunic D. Terminal Synthesis v2. MSDS Programme (2026).

[2] Kadunic D. Paper O: Scale Factor. MSDS Tales of the Omega (2026).

[3] Kadunic D. Series N v2.2. doi:10.5281/zenodo.19476384.

[4] Watkins R., Feldman H.A., Hudson M.J. MNRAS 392, 743 (2009). Bulk flow.

[5] Tully R.B. et al. Astron.J. 152, 50 (2016). CosmicFlows-4.

[6] EUCLID Collaboration. A&A; 642, A191 (2020). Science case.

[7] DESI Collaboration. arXiv:2404.03002 (2024). Dark energy.

[8] Desjacques V., Jeong D., Schmidt F. Phys.Rept. 733, 1 (2018). LSS review.